Colorectal cancer RNA-seq: tumour vs normal, metastasis vs primary

Analysis from raw reads, public dataset GSE50760 (Kim et al., 2015)

Summary

Paired RNA-seq from colorectal cancer patients (normal colon, primary tumour and liver metastasis from the same person), processed from FASTQ files to pathways. Tumour vs matched normal gives the expected biology: 836 differentially expressed genes, all 10 reference markers moving the right way, and a clear proliferation signature (E2F and MYC targets, G2-M checkpoint). Metastasis vs primary is dominated by liver genes coming from host tissue in the sample, so I treat that contrast as a tissue-contamination result and not as tumour biology.

Data and methods

I used 6 of the 18 patients who have all three tissues (18 samples) and the first 2M read pairs of each sample, to fit a free Colab session. Reads were trimmed with fastp and quantified with Salmon against GENCODE v46, then summed to gene level. After filtering (at least 10 reads in 6 or more samples) 12,336 genes remained. Differential expression used DESeq2 (pydeseq2) with the design ~ *patient* + *tissue*; genes count as significant at padj < 0.05 and |log2FC| > 1. Pathways: Enrichr (GO Biological Process) on the top 300 genes per direction, and GSEA on the MSigDB Hallmark sets using all genes.

Quality was uniform across samples: 90-92% of reads mapped to the transcriptome, and Q30 after trimming was 85-90%.

Sample structure

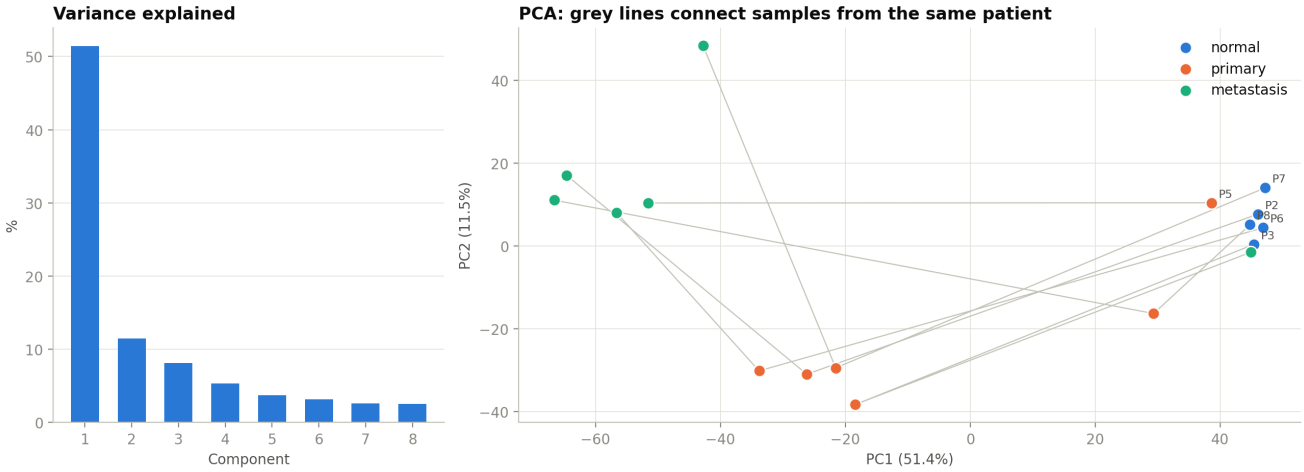


Figure 1. Variance per component (left) and PC1 vs PC2 (right). Grey lines join the three samples of each patient.

PC1 and PC2 (51% and 12% of variance) separate the tissues (eta-squared 0.61 and 0.60 for tissue). PC3 to PC5 follow the patient instead (0.64-0.69). A large part of the variation is therefore between people, which is why patient is in the model.

Tumour vs normal

836 genes pass the thresholds: 402 up and 434 down in tumour. Among the strongest are ETV4, SLC6A20, OLFM4 and STC2, all up. Markers of normal colon epithelium fall in tumour (AQP8, CA1, GUCA2B, GUCA2A, SLC26A3, CA2), while MMP7, CLDN1, MYC and LGR5 rise. All 10 go in the expected direction. With 6 patients not all are individually significant: 6 of the 10 pass padj < 0.05 (LGR5, MMP7, GUCA2A and GUCA2B do not).

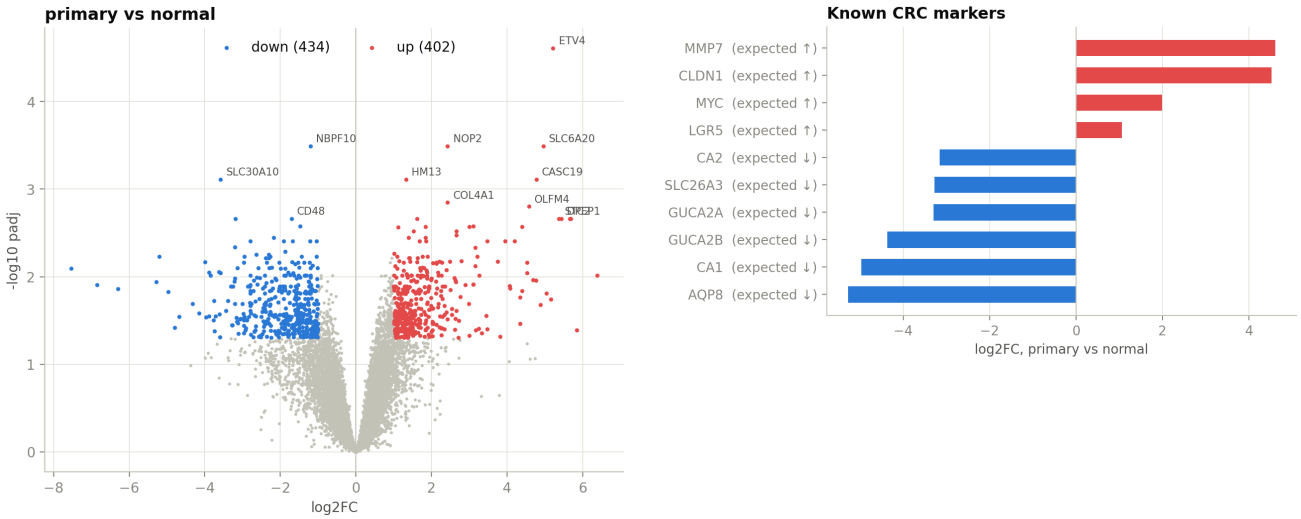


Figure 2. Volcano plot for primary vs normal (left) and log2 fold change of the reference markers (right).

Pathways

Up in tumour: E2F targets, MYC targets (V1, V2), G2-M checkpoint, unfolded protein response, mTORC1 signalling and DNA repair. Down: inflammatory and interferon gamma response, oxidative phosphorylation, fatty acid metabolism and adipogenesis. GO on the top genes agrees: upregulated genes are enriched for ribosome biogenesis, DNA replication and mitotic cell cycle; downregulated genes for MHC class II antigen presentation and the response to zinc ion. The drop in immune terms may partly reflect different immune cell content in tumour and normal mucosa; this analysis cannot separate that from expression changes within cells.

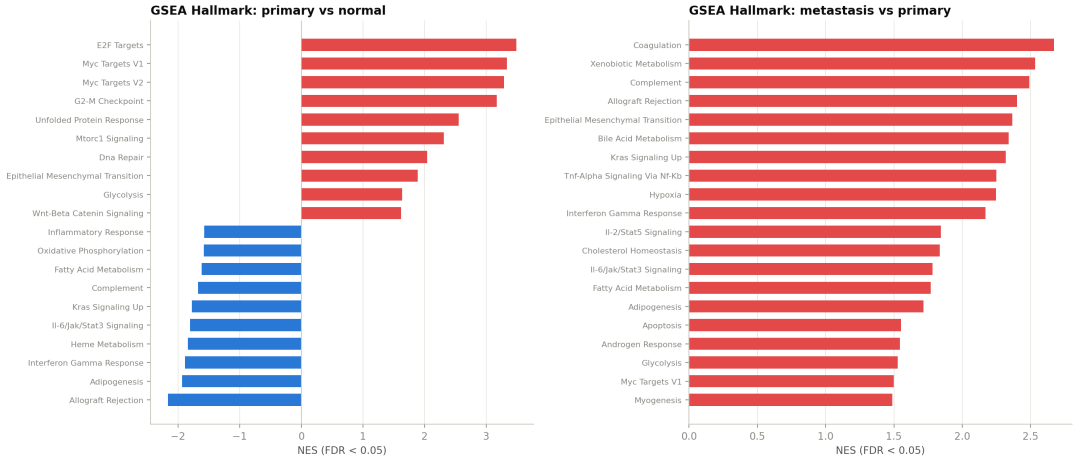


Figure 3. GSEA on Hallmark gene sets, FDR < 0.05. Positive NES means higher in the first group of the comparison.

Metastasis vs primary

215 genes pass (179 up, 36 down). The top hits are liver genes: FGA, FGB, FGG, HP, ALB, KNG1, CP and GATM, and the Hallmark results (coagulation, complement, xenobiotic and bile acid metabolism) point the same way. That is host hepatocyte RNA in the metastasis samples, not a change in the tumour cells. EMT and KRAS signalling also come up in this contrast, but with this much liver signal I would not read them as tumour-intrinsic without deconvolution or a tumour-purity correction.

Limitations

Only 6 of 18 patients and a 2M read-pair subsample per sample, so small effects and weakly expressed genes are underpowered. The Salmon index has no decoys, and gene counts are summed NumReads without the length correction tximport applies. Tumour samples contain stroma and immune cells and no deconvolution was done. The results show the workflow and its sanity checks; they are not new biology.

Alongside this report: the Colab notebook (FASTQ to figures), differential expression tables for both contrasts, normalised counts and the MultiQC report.